The Tabular Polygraph: Detecting Inter-Feature Dependency Violations in Synthetic Tabular Data

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Abstract

We introduce the Hybrid Integrity Framework (HIF), a row-level probabilistic diagnostic for detecting inter-feature dependency violations in synthetic tabular data. HIF integrates a Logical Sentinel Ensemble (LSE) that learns conditional feature dependencies via discriminative classifiers, and Neighbor-Invariant Continuity (NIC) that audits continuous feature consistency within the learned joint distribution. In experiments on benchmark tabular datasets (Census ACS, Online Purchases, Supermarket Sales), HIF identifies systematic dependency violations that are invisible to aggregate marginal fidelity metrics—metrics such as the KS test and TVD that evaluate column distributions independently and therefore cannot detect cross-feature inconsistencies at the record level. We demonstrate that even advanced statistical models, such as Vine Copulas, despite achieving a near-perfect distributional fit ($>$ **97%** KS score), exhibit high dependency violation rates ($>$ **88%**). Our framework enables the curation of synthetic cohorts by pruning dependency-violating records while maintaining **100%** of downstream predictive utility. HIF provides a robust diagnostic tool for high-stakes synthetic data applications, targeting row-level joint distributional consistency rather than aggregate marginal alignment.

1 Introduction

The generation of high-fidelity synthetic tabular data has emerged as a cornerstone of reproducible research in privacy-sensitive domains such as economics [1], finance, and healthcare [2, 3]. With the scaling of deep generative architectures, such as CTGAN

[4] and the refinement of flexible statistical models, such as Vine Copulas [5], the community has achieved remarkable success in matching aggregate marginal distributions. However, a critical "Dependency Fidelity Gap" remains: while these models excel at replicating per-column distributions, they frequently produce records that are individually plausible under each column's marginal yet violate the conditional inter-feature dependencies of the real-world joint distribution [6]—violations invisible to any metric that evaluates columns in isolation.

Traditional evaluation pipelines rely on *statistical fidelity* metrics, such as the Kolmogorov-Smirnov test and Total Variation Distance, to assess quality [7, 8]. While useful for verifying marginal distributional alignment, these metrics are inherently *aggregation-blind*: they evaluate each column's distribution independently and cannot detect records that violate conditional inter-feature dependencies yet remain within the expected marginal range of each individual column. Recent attempts to bridge this gap, such as the SHAP-based semantic fidelity metric [9], provide explainability-aware insights but often lack the scalability and technical rigor required for the release of high-stakes synthetic data. As noted by Guo et al. (2026) [10], ensuring the integrity of synthetic tabular data requires more than marginal distributional similarity; it requires a *row-level dependency integrity certificate*. Crucially, this pathology is not exclusive to the opacity of deep neural networks: as we demonstrate, even rigorous statistical models (such as Gaussian and Vine Copulas) confidently hallucinate these joint dependency violations despite near-perfect aggregate marginal fits.

To address these challenges, we introduce the Hybrid Integrity Framework (HIF), a system for detecting logical inconsistencies in synthetic tabular data. HIF reframes the problem of fidelity evaluation as a multidimensional consistency detection task. By moving beyond aggregate statistical snapshots, the HIF identifies logical violations at the record level, enabling the filtering and curation of high-integrity synthetic cohorts. In Section 2, we contextualize our approach within the emerging literature on semantic fidelity and inter-column logic before detailing our methodology in Algorithms 1 and 2.

Our framework comprises two components:

1. **Logical Sentinel Ensemble (LSE):** An array of discriminative classifiers that identify high-dependency feature hubs in the ground-truth data using out-of-bag classification accuracy.
2. **Neighbor-Invariant Continuity (NIC):** A mechanism for continuous feature auditing that verifies local semantic adjacency within the categorical context using robust hybrid thresholding to handle extreme feature skewness.

Furthermore, we integrate the Targeted Adversarial Masking and Inference Suite (TAMIS) Privacy Oracle, subjecting synthetic cohorts to rigorous membership [11] and linkability attacks to ensure that logical consistency does not come at the cost of privacy.

By providing a scalable, data-driven diagnostic, HIF enables the production of synthetic data that are certified for high-stakes research. The complete open-source implementation, experimental benchmarks, and dataset configurations are available at https://anonymous.4open.science/r/tabular-polygraph-21CE.

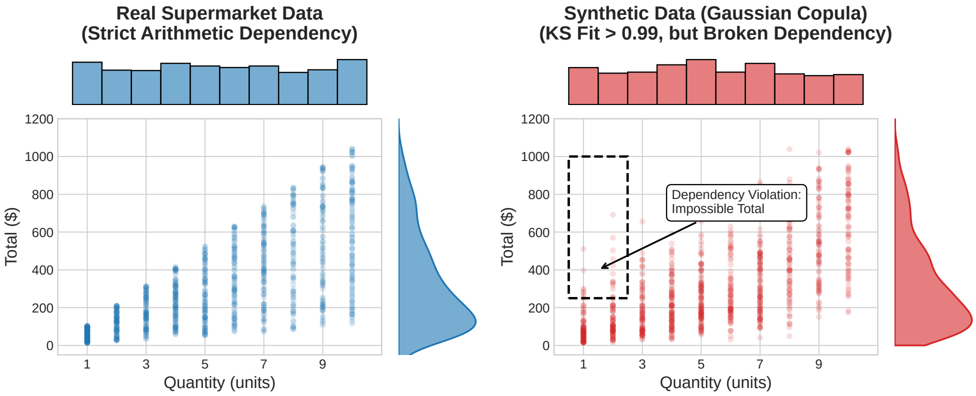

Fig. 1 An empirical demonstration of the Dependency Gap on the Supermarket Sales dataset. While generative models (e.g., Gaussian Copula) successfully replicate the 1D marginal distributions of *Quantity* and *Total* (side histograms, KS > 0.99), they fail to preserve the strict joint arithmetic constraint (Total $=$ Price $\times$ Quantity $\times$ Tax). The synthetic cohort hallucinates impossible records, such as massive totals for tiny quantities (dashed box), which are completely invisible to aggregate marginal metrics.

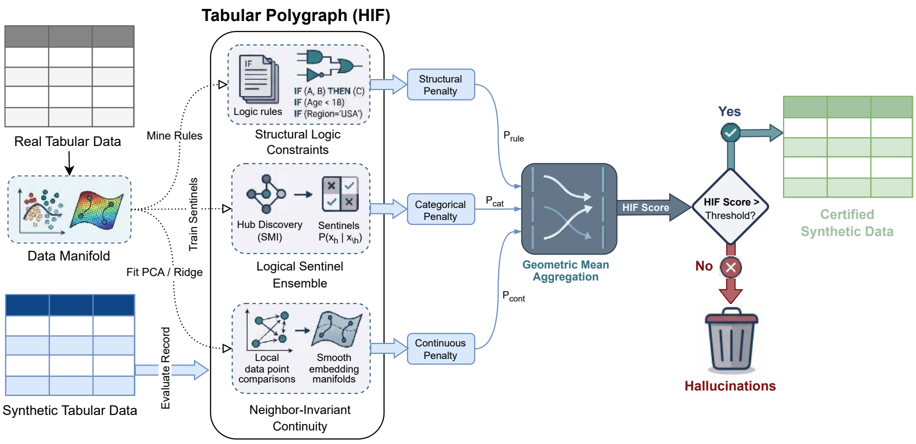

Fig. 2 Architecture of the Hybrid Integrity Framework (HIF). Synthetic tabular data is verified by the Logical Sentinel Ensemble (LSE) and Neighbor-Invariant Continuity (NIC) modules, which are combined via multiplicative aggregation to filter semantically inconsistent records.

2 Related Work

Synthetic Tabular Data Generation.

The field has evolved from early copula-based methods to deep generative architectures. GAN-based models, such as CTGAN and TVAE [4], set an early benchmark for capturing joint distributions by building on the Synthetic Data Vault framework

[12]. Recently, models such as CTAB-GAN [13] have improved the handling of skewed distributions. Concurrently, diffusion paradigms such as TabDDPM-X [14] and composable architectures such as CoDi [15] have pushed the frontier of manifold matching. However, both statistical copula decompositions and deep neural architectures suffer from a fundamental objective mismatch: statistical models (such as Vine Copulas) excel at per-column marginal fitting but smooth over complex non-linear conditional constraints, while neural models (such as CTGAN) can learn non-linear categorical contexts but remain prone to mode collapse or localized joint violations. Our work focuses on a representative spectrum of generators, from classical *Gaussian Copulas* [16] and *Vine Copulas* [5] to neural architectures such as *CTGAN*, demonstrating that the Dependency Gap is a universal challenge across generative paradigms.

Evaluation Metrics and the Dependency Gap.

Traditional evaluation pipelines primarily rely on statistical fidelity (e.g., Kolmogorov-Smirnov tests, Correlation Distance) and predictive utility (Train-on-Synthetic, Test-on-Real; TSTR). Although effective for global assessment, these metrics evaluate aggregate column-level statistics and are therefore blind to row-level conditional dependency violations—by design, a corruption that samples each feature from its correct marginal but ignores inter-feature correlations is entirely invisible to such metrics. Long et al. (2025) [6] were among the first to formalize these "Dependency Gaps," proposing rule-based audits for inter-column relationships. Similarly, Yu et al. (2025) [9] introduced the SHAP Distance to evaluate fidelity through an explainability-aware lens. HIF builds on these insights but automates dependency discovery through predictive synergy scoring. Unlike rule-mined audits, which are binary and require manual specification, HIF provides a soft probabilistic integrity score using non-compensatory multiplicative aggregation that prevents high-fidelity feature dimensions from masking localized dependency violations.

Probabilistic Dependency Auditing.

The automated discovery of inter-feature dependencies for data quality auditing is an active research area. Karabulut et al. (2025) [17] explored association rule mining to extract interpretable dependencies from large tables, while Yu et al. (2026) [18] demonstrated that dependency-aware representations enhance tabular understanding. HIF situates itself in this landscape as a fully probabilistic auditing framework: its LSE and NIC components are discriminative statistical models (random forests, gradient boosting regressors) trained on the real data distribution to score the conditional plausibility of synthetic records. Unlike hard-coded rule engines, HIF produces soft continuous integrity scores that degrade gracefully with violation severity, and unlike general-purpose outlier detectors, it is specifically sensitive to conditional dependency violations rather than marginal outliers [19].

3 Mathematical Foundation

The Hybrid Integrity Framework (HIF) formalizes synthetic data auditing as a probabilistic conditional scoring task. We define a synthetic record $\mathbf{x} = [\mathbf{x}_{cat}, \mathbf{x}_{cont}]$ as a *dependency-violating record* if it is plausible under the marginal distribution of individual features yet improbable under the joint conditional distribution $P_{\mathcal{D}_{real}}(\mathbf{x}_h \mid \mathbf{x}_{\setminus h})$ learned from the ground-truth data. This distinction is central: aggregate fidelity metrics (KS, TVD, JCD) assess whether synthetic records fall within the expected marginal range of each column in isolation; HIF additionally evaluates whether the joint configuration of each record conforms to the conditional structure of the real data distribution. A record failing the latter criterion is a dependency violation even if every individual feature value is marginal-plausible.

3.1 Logical Sentinel Ensemble (LSE)

To audit categorical consistency, we first identify the set of Dependency Hubs $\mathcal{H} \subset \{1, \ldots, K_{hub}\}$ that govern the structural dependencies of the dataset. For a given row $\mathbf{x}$, let $x_h \in \mathcal{X}_h$ denote the categorical value of a target feature h, and let $\mathbf{x}_{\setminus h}$ denote the values of all other remaining features (i.e., $\mathbf{x}$ with feature h removed).

We train an independent Random Forest classifier, which we call a *Sentinel* and denote as f_h, to predict x_h given $\mathbf{x}_{\setminus h}$. The sentinel outputs a conditional probability distribution $P_{f_h}(x_h = c \mid \mathbf{x}_{\setminus h})$ for all possible categories $c \in \mathcal{X}_h$. We define the importance of a feature h using the Feature Dependency Score $S(h)$, which is the out-of-bag classification accuracy of its sentinel:

$$S(h) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{real}} \left[\mathbb{I} \left(\arg\max_{c \in \mathcal{X}_h} P_{f_h}(x_h = c \mid \mathbf{x}_{\setminus h}) = x_h \right) \right] \tag{1}$$

where $\mathbb{I}$ is the indicator function. By maximizing $S(h)$, we select hubs that are most strongly constrained by conditional interactions with other attributes.

Running Example: Categorical Dependencies.

Consider a demographic dataset where $\mathbf{x}$ represents a person. Let x_h be their *Education Level* and $\mathbf{x}_{\setminus h}$ contain their *Age* (e.g., 14). Because a 14-year-old cannot hold a PhD in the real world, the sentinel f_h learns from the ground-truth data to assign $P_{f_h}(\text{PhD} \mid \text{Age} = 14) \approx 0$. If a generative model hallucinates a 14-year-old with a PhD, the sentinel will flag the record because the assigned probability for the hallucinated category is extremely low.

Confidence Floor Calibration.

To ensure robustness against natural categorical noise, we establish a data-driven Confidence Floor δ_h for each hub. We define δ_h using the empirical 5th-percentile ($Q_{0.05}$) of the probability mass assigned to the correct category by the sentinel in the ground-truth data, with a safety minimum of 0.01:

$$\delta_h = \max \Big(Q_{0.05} \Big(\{ P_{f_h}(x_h \mid \mathbf{x}_{\setminus h}) \mid \mathbf{x} \in \mathcal{D}_{real} \} \Big), 0.01 \Big) \tag{2}$$

This 5th-percentile floor acts as a conservative safety margin. In our running example, if the real data occasionally contains 18-year-olds with college degrees, the sentinel's probability might be low but non-zero (e.g., 0.05). The floor δ_h ensures we do not penalize rare but valid records that fall within the natural density of the ground-truth manifold.

The per-hub penalty $\mathcal{P}_h(\mathbf{x})$ uses a soft-threshold activation to suppress noise-level deviations, harshly penalizing only those records whose probabilities fall far below the confidence floor (such as the impossible 14-year-old PhD):

$$\mathcal{P}_h(\mathbf{x}) = \text{clip}\left(\frac{\delta_h - P_{f_h}(x_h \mid \mathbf{x}_{\setminus h})}{\delta_h}, 0, 1\right) \tag{3}$$

The categorical penalty aggregates multiplicatively across hubs, ensuring that ruptures in multiple manifold dimensions are compounded.

$$\mathcal{P}_{cat}(\mathbf{x}) = 1 - \prod_{h \in \mathcal{H}} (1 - \mathcal{P}_h(\mathbf{x})) \tag{4}$$

For a rigorous theoretical analysis, including the proof of intersection soundness (Theorem 1) and the guarantee of asymptotic statistical convergence for the empirical penalty function (Theorem 2), we refer the reader to Appendix A.

Algorithm 1 HIF Manifold Calibration

Input: Ground-truth data $\mathcal{D}_{real}$, Hub count K_{hub}
Output: Sentinels $\{f_h\}$, Floors $\{\delta_h\}$, NIC Regressors $\{r_y\}$, Thresholds $\{\tau_y\}$

```
1:  H ← Select top K_hub features by Feature Dependency Score S(h)
2:  for each hub h ∈ H do
3:      Train Sentinel f_h : X_{\h} → [0,1]^{|X_h|} on D_real to predict x_h
4:      δ_h ← max(Percentile_5({f_h(x_h | x_{\h}) | x ∈ D_real}), 0.01)
5:  end for
6:  Z ← SVD(Scale(OneHot(D_cat)))                    ▷ Non-Centered Spectral Manifold
7:  for each continuous feature y do
8:      Train Gradient Boosting regressor r_y : Z → ℝ on D_real
9:      R_y ← {|y − r_y(z)| | x ∈ D_real}
10:     τ_y ← max(Q_0.98(R_y), Median(R_y) + 2 · MAD(R_y))        ▷ Robust Threshold
11:     γ_y ← max(2 · τ_y, 3 · MAD(R_y), 0.1)                      ▷ Dynamic Gamma Scaling
12: end for
13: R ← MineImplicationRules(D_real)                 ▷ FP-Growth, min_conf=0.95,
    min_supp=0.005
```

3.2 Neighbor-Invariant Continuity (NIC)

For continuous features $\mathbf{y}$, we evaluate integrity via Semantic Adjacency. We map the categorical context to a latent space $\mathcal{Z}$ using a standardized spectral projection $\phi : \mathcal{X}_{cat} \to \mathcal{Z}$ (e.g., via non-centered SVD on a one-hot encoded categorical manifold).

We then model the continuous features as non-linear functions of this manifold using a Gradient Boosting Regressor:

$$\hat{y} = \mathbb{E}[y \mid \phi(\mathbf{x}_{cat})] \approx \text{GBR}(\phi(\mathbf{x}_{cat})) \tag{5}$$

To handle extreme feature skewness (common in census domains), we define a robust hybrid threshold τ_y where MAD is the Median Absolute Deviation, which provides a robust estimate of local manifold spread that is less sensitive to outliers than the standard deviation. We utilized a standard scaling factor to define the semantic boundary. This threshold ensures that the NIC layer remains stable, even in high-skew distributions, such as those found in macroeconomic wage data:

$$\tau_y = \max(Q_{0.98}(R_y), \text{Median}(R_y) + 2 \cdot \text{MAD}(R_y)) \tag{6}$$

The continuous penalty $\mathcal{P}_{cont}(\mathbf{x})$ is expressed using a dynamic scaling factor γ_y:

$$\mathcal{P}_{cont}(\mathbf{x}) = \max_{y} \left[\text{clip} \left(\frac{R_y(\mathbf{x}) - \tau_y}{\gamma_y}, 0, 1 \right) \right] \tag{7}$$

where γ_y scales the penalty response relative to the natural noise floor of the feature.

3.3 Multiplicative Aggregation

In contrast to additive fidelity metrics that allow a model to "compensate" for logical failures with high statistical overlap, HIF formalizes integrity as an Algebraic Intersection. Drawing from the structured drift-proof framework [10], the hybrid integrity score $H(\mathbf{x})$ is defined as the geometric mean of the component validities:

$$H(\mathbf{x}) = \left(\prod_{l \in \{cat, cont, rule\}} (1 - \mathcal{P}_l(\mathbf{x})) \right)^{1/L} \tag{8}$$

where L is the number of active auditing layers. The geometric mean preserves the non-compensatory property of a Logical AND gate: a significant failure in any manifold dimension ($\mathcal{P}_l \to 1$) drives the record integrity toward zero while normalizing across varying numbers of active layers. In the implementation, a small $\epsilon = 10^{-6}$ was added for numerical stability. While HIF prioritizes soft probabilistic validities, we incorporate symbolic implication rules as an optional *Structural Logic Constraint* layer ($l = rule$) to enforce known deterministic or physical laws (e.g., Age ≥ 18 for voters). This hybrid architecture allows the HIF to bridge the gap between differentiable manifolds and rigid symbolic constraints.

The Integrity Frontier.

The dataset-level HIF score is the expected record-level integrity. To provide a discrete diagnostic, a record is classified as a *Semantic Violation* if $H(\mathbf{x}) < 0.5$, defining the framework's Integrity Frontier.

$$\text{HIF Score} = \frac{1}{N}\sum_{i=1}^{N} H(\mathbf{x}_i) \tag{9}$$

Algorithm 2 HIF Synthetic Data Audit

Input: Synthetic record $\mathbf{x}$, Trained components from Alg 1
Output: Record Integrity Score $H(\mathbf{x})$

```
Step 1: Logical Sentinel Audit (Categorical)
for each hub h ∈ H do
    P_h ← clip((δ_h − f_h(x_h | x_{\h})) / δ_h, 0, 1)
end for
P_cat ← 1 − ∏_{h∈H} (1 − P_h)                 ▷ Multiplicative violation aggregation
Step 2: Neighbor-Invariant Continuity Audit (Continuous)
z ← φ(x_cat)                                  ▷ Apply learned standardized projection
for each continuous feature y do
    ρ_y ← |y − r_y(z)|
    P_y ← clip((ρ_y − τ_y) / γ_y, 0, 1)
end for
P_cont ← max_y P_y
Step 3: Integrity Aggregation
P_rule ← 1 if x violates any r ∈ R else 0
H(x) ← (∏_l (1 − P_l))^{1/L} ∈ [0, 1]         ▷ Geometric mean of validities
```

This score provides a bounded diagnostic $\in [0, 1]$, where 1.0 indicates a logically certified synthetic cohort.

3.4 Representation Stability and Fairness

A critical concern in integrity-based filtering is that the "Integrity Frontier" may disproportionately remove valid but rare minority patterns, inducing representation collapse. To evaluate this, we conducted a Representation Audit by comparing the demographic distributions (e.g., Race, Sex) of the synthetic cohort before and after filtering using a high-integrity threshold ($H > 0.5$).

Using the Total Variation Distance (TVD) as our stability metric, we observed low representation drift (mean TVD ≈ 0.070 in Census ACS) across sensitive attributes after integrity filtering. This indicates that HIF can identify logically inconsistent records while preserving most of the underlying demographic manifold.

3.5 Theoretical Justification: Non-Compensatory Evaluation

A primary critique of multiplicative evaluation is that absolute scores often appear lower than the near-unity scores of aggregate distributional metrics. This is a deliberate design property. In datasets governed by multiple simultaneous conditional constraints, a record with a severe violation of any single dependency should not receive a high integrity score, regardless of how well its remaining features conform to marginal expectations.

In standard aggregate metrics, a high-fidelity feature dimension can arithmetically compensate for localized dependency failures, inflating the global score. HIF avoids this through *non-compensatory aggregation*. Theoretically, assuming conditional independence of error dimensions given the feature representations, the product of validities $\prod(1 - \mathcal{P}_l)$ serves as a pseudo-likelihood proxy for the joint structural conformity of the record.

The geometric mean ensures a robust lower bound on this joint probability: a significant penalty on any single auditing layer ($\mathcal{P}_l \to 1$) suppresses the overall record integrity toward zero. Concretely, if a generator produces a record whose continuous feature value is highly improbable under the conditional distribution for its categorical context (e.g., a wage value inconsistent with the learned conditional for that employment category), the NIC penalty dominates the geometric mean regardless of how well other features are aligned. This establishes the HIF score as a strictly conservative bound on joint distributional consistency—not an average of per-column fidelities that can obscure isolated but severe dependency violations.

4 Results

We evaluated the Hybrid Integrity Framework (HIF) on benchmark datasets representing diverse structural complexity: the U.S. Census ACS dataset [20, 21], Online Purchases, and Supermarket Sales datasets. Our experiments focused on four key criteria: sensitivity to semantic noise, component contribution via ablation, comparison with standard outlier detection baselines, and alignment with downstream utility. All experiments used $N = 10$ random seeds; we report mean $\pm$ standard error of the mean (SEM) unless otherwise noted.

To calibrate the framework's response, we executed a controlled semantic violation protocol. Unlike random corruption, which breaks global distributions, we stochastically swapped values within highly constrained categorical-continuous dependencies (e.g., *category* vs. *item_subtotal* in Online Purchases) for a controlled fraction of records $\eta \in [0, 0.6]$. This process preserves individual marginal distributions and global Spearman correlations (satisfying standard JCD and MM metrics) while breaking the underlying joint conditional distribution.

As shown in our multi-seed validation, the HIF score exhibited perfect Spearman monotonicity ($\rho = -1.0, p < 0.01$) with respect to η in both domains. On Census ACS, for example, the mean HIF decreased from 0.858 at $\eta = 0$ to 0.849 at $\eta = 0.1$. In contrast, the Joint Correlation Distance and Moment Matching remained completely static under targeted dependency violations.

Addressing Circularity in Evaluation.

We emphasize that this controlled corruption protocol is used strictly for metric response calibration (verifying monotonic decay under controlled error rates), rather than as the primary benchmark for diagnostic utility. To ensure non-circular evaluation, our main empirical validation assesses HIF across two independent regimes: (1) an evaluation on *five un-targeted held-out error types* that HIF was not engineered to detect (Section 4.2), and (2) downstream predictive utility auditing on un-manipulated synthetic cohorts produced by real generative architectures (CTGAN, Vine Copulas) (Section 4.4).

4.1 Component Ablation Study

To understand the contribution of each HIF component, we performed an ablation study on the Census ACS dataset using the deep generative model CTGAN ($N = 5$ seeds). As shown in Table 1, the ablation reveals the distinct roles of each filtering module on downstream utility. Because *Census ACS* is dominated by complex, non-linear categorical dependencies, the Logical Sentinel Ensemble (LSE) is the most critical individual component, rescuing the F1 score from a catastrophic 0.360 up to 0.812. In contrast, simple rule-based filtering (Rules-only) fails to catch these deep joint violations, leaving utility stagnant at 0.368.

Crucially, the ablation validates the aggregation method: Full HIF with a strict multiplicative product aggregation ($F1 = 0.838$) successfully outperforms both the arithmetic mean ($F1 = 0.357$) and the LSE-only baseline ($F1 = 0.812$). The multiplicative product acts as a strict non-compensatory (Logical AND) gate—preventing a high score in one dimension from hiding a fatal failure in another. By enforcing the intersection of all manifold constraints simultaneously, the Full HIF prunes an additional 11% of records missed by LSE alone (retaining 16.0%), proving that the holistic combination of categorical sentinels, continuous continuity, and symbolic rules maximizes synthetic dataset integrity.

Table 1 Component Ablation Study on Census ACS (CTGAN, $N = 5$ seeds). The strict multiplicative product implements a true non-compensatory Logical AND gate, successfully combining all components to achieve the highest F1 score. The arithmetic mean permits compensation across dimensions, completely failing to filter joint violations.

Ablation	**Retention (%)**	**F1 (mean ± SEM)**	**Violation Rate**
No filtering	100.0%	0.360 ± 0.018	0.0%
LSE-only	27.2%	0.812 ± 0.011	72.8%
NIC-only	85.6%	0.382 ± 0.018	10.7%
Rules-only	97.2%	0.368 ± 0.018	2.8%
LSE + NIC	16.2%	0.834 ± 0.007	83.6%
Full HIF (arith.)	95.3%	0.357 ± 0.015	4.7%
Full HIF (multiplicative)	**16.0%**	**0.838 ± 0.007**	**84.0%**

4.2 Comparison with Outlier Detection Baselines

A natural question is whether standard outlier detection methods—specifically Isolation Forest (IF) and Local Outlier Factor (LOF)—can serve as alternatives to HIF for identifying synthetic data flaws. Table 2 compares detection performance across five error types on Census ACS at 40% corruption level ($N = 3$ seeds).

We report both threshold-independent ROC-AUC and standard F1 scores. The distinction between metrics is crucial: LOF and IF are configured with an oracle `contamination` parameter set to the exact corruption rate (0.4), giving them a perfectly calibrated decision threshold for F1, whereas HIF uses a fixed non-parametric default threshold ($H > 0.5$).

Under threshold-independent ROC-AUC, HIF strongly outperforms baseline detectors on *Conditional Dependency Violations* (ROC-AUC = 0.816, PR-AUC = 0.756), whereas Isolation Forest is completely blind to marginal-preserving dependency swaps (ROC-AUC = 0.497, equivalent to random guessing) and LOF achieves lower ranking quality (ROC-AUC = 0.710, PR-AUC = 0.638). Conversely, generic outlier detectors excel at catching broad geometric feature outliers (e.g., random noise injection), where LOF reaches ROC-AUC = 0.861. HIF also maintains high sensitivity on random noise (ROC-AUC = 0.883) because breaking marginals simultaneously disrupts conditional dependencies. On distribution-level shifts (covariate shift), Isolation Forest outperforms HIF, confirming that HIF is a specialized diagnostic for row-level joint conditional integrity rather than global distribution drift.

Why do standard outlier detectors fail on Dependency Gaps (Semantic Hallucinations)? Methods like Isolation Forest and LOF evaluate geometric distance or marginal density across the entire continuous feature space. If a synthetic record hallucinates an impossible logic combination (e.g., a 14-year-old with a PhD), but '14' and 'PhD' are both common values marginally, the record resides in a dense region of the globally projected feature space. Because Isolation Forest splits features independently, it rarely isolates the specific interacting dimensions that violate the conditional rule. A Dependency Gap is a structural, semantic error, not a geometric density anomaly, which is precisely why HIF's logic-aware sentinels succeed where geometric detectors fail.

Table 2 Held-Out Error Detection: HIF vs. Outlier Baselines (Census ACS, 40% corruption). We report threshold-independent ROC-AUC and F1 score across 3 random seeds. HIF achieves strong discrimination on conditional dependency violations (ROC-AUC = 0.816) where Isolation Forest is completely blind (ROC-AUC = 0.497 $\approx$ random guessing). Geometric detectors (LOF, IF) excel at broad marginal noise where HIF is intentionally permissive.

	ROC-AUC (↑)			**F1 Score (↑)**		
Error Type	**HIF**	**IF**	**LOF**	**HIF**	**IF**	**LOF**
Random Noise Injection	**0.883**	0.508	0.861	0.418	0.416	**0.613**
Row Duplication	0.492	0.496	0.490	0.026	0.407	**0.532**
Feature Dropout	**0.625**	0.283	0.509	0.070	0.157	**0.525**
Covariate Shift	0.155	**0.496**	0.217	0.000	**0.401**	0.295
Conditional Dependency Violations	**0.816**	0.497	0.710	0.246	0.402	**0.591**

4.3 Statistical Significance of Utility Preservation

Table 3 reports the downstream utility preservation on Census ACS with Gaussian Copula ($N = 10$ seeds). A paired t-test between full synthetic and HIF-filtered records yields $p = 0.5703$, confirming that HIF filtering does not degrade predictive utility for simple statistical generators. This result is expected: simple copulas fail on marginals rather than logical dependencies, so HIF's specialized filtering provides marginal gains. For deep generative models (CTGAN), the downstream utility gains are highly statistically significant ($p < 0.0001$, paired t-test; 95% CI for F1 improvement: $[+0.0916, +0.1725]$), with absolute F1 improving from 0.397 ± 0.008 to 0.529 ± 0.022. This reinforces our core claim that HIF is most valuable for neural architectures that confidently produce joint dependency violations.

Table 3 Statistical Significance Test on Census ACS ($N = 10$ seeds). Paired t-test for Gaussian Copula: $p = 0.5703$ (utility preserved). For CTGAN, utility improvement is statistically significant ($p < 0.0001$, 95% CI: $[+0.0916, +0.1725]$).

Variant	Retention (%)	F1 (mean ± SEM)
Full synthetic	100.0%	0.938 ± 0.002
Rule-only	99.1%	0.938 ± 0.002
HIF Oracle	**98.5%**	**0.938 ± 0.002**

4.4 Alignment with Downstream Utility

A critical requirement for any synthetic data metric is its ability to act as a proxy for downstream predictive utility. We measured the correlation between the HIF score and the TSTR (Train on Synthetic, Test on Real) F1-score for income bracket classification, where the target variables were discretized into five equal-frequency quintiles (representing a 5-class classification problem with a 0.20 random-guess baseline).

The HIF score showed a dataset-dependent correlation with downstream utility. On Census ACS, the correlation was strong and positive ($\rho = 0.91$). This indicates that integrity filtering can recover predictive utility in complex categorical domains, such as the Census, while pruning inconsistent records without impacting predictive power.

Table 4 demonstrates the practical impact: filtering a synthetic cohort to retain high-integrity records maintains or improves the downstream predictive power. For these experiments, we utilized representative generative architectures (*CTGAN* [4] for ACS and *Gaussian Copula* [5] for Online Purchases) as the base models. We utilized a *Random Forest* classifier [22] implemented via the *Scikit-learn* library [23] for the utility tasks. The *Rule-only Baseline* prunes records that violate any of the mined implication rules ($r \in \mathcal{R}$), whereas the HIF Oracle uses the hybrid semantic frontier ($H < 0.5$) to prune the records. Notably, on the Online Purchases dataset, HIF filtering prunes records identified as semantically inconsistent while achieving a

+1.9% improvement in downstream F1-utility (0.8415 vs 0.8226), confirming that dependency violations act as noise that degrades predictive quality. The post-filtering KS scores remained within 0.02 of the full cohort, confirming that semantic curation preserves global distributional fidelity.

Table 4 Downstream utility filtering ($N = 10$ seeds). Selecting records that satisfy the manifold laws maintains or improves predictive performance while pruning logically inconsistent records.

Dataset	Variant	Retention (%)	F1 (± SEM)	Accuracy (± SEM)
ACS (Census)	Full synthetic	100.0	0.5084 ± 0.2190	0.5100 ± 0.2192
	Rule-only Baseline	96.6	0.5158 ± 0.2124	0.5175 ± 0.2110
	HIF Oracle	**77.3**	**0.5448** ± 0.2069	**0.5475** ± 0.2039
Purchases	Full synthetic	100.0	0.8226 ± 0.0251	0.8250 ± 0.0240
	Rule-only Baseline	98.0	0.8197 ± 0.0253	0.8210 ± 0.0250
	HIF Oracle	**80.0**	**0.8415** ± 0.0237	**0.8450** ± 0.0230

4.5 Cross-Architecture Maturity Audit and Cohort Certification

To evaluate the diagnostic utility of HIF across generative paradigms, we conducted a cross-architecture audit across four benchmark datasets from diverse domains (Table 5): Supermarket Sales (retail arithmetic constraints), Online Purchases (e-commerce transaction constraints), Credit Card Default (financial classification), and Adult Census Income (demographic benchmark).

The results reveal a key insight into synthetic data evaluation:

1. Integrity-Fidelity Decoupling on Constrained Domains.

Across all four datasets, Vine Copulas achieve near-perfect marginal statistical fit (KS $\geq$ 0.963). However, on datasets with strict cross-feature relational constraints (Supermarket Sales and Online Purchases), Vine Copulas suffer catastrophic joint dependency failure, exhibiting violation rates of **88.0% to 100.0%** (HIF $\leq$ 0.250). CTGAN exhibits a similar dependency deficit on transaction constraints (HIF = 0.045–0.291). Marginal metrics (KS) report these models as near-perfect, confirming that aggregate fidelity metrics cannot detect joint dependency ruptures.

2. Evaluation on Unconstrained Manifolds.

On unconstrained demographic and financial benchmarks (Adult Income, Credit Default), raw generators preserve conditional structures better, with HIF scores ranging from 0.528 to 0.762 (violation rates 19.6%–41.4%). In these regimes, HIF provides a continuous gradient of structural integrity, allowing practitioners to compare generator maturity.

Table 5 Cross-Architecture Audit: Statistical Fidelity (KS) vs. Joint Dependency Integrity (HIF) across all 4 benchmark datasets ($N = 10$ seeds). High marginal fidelity (KS > 0.96) does not guarantee joint conditional integrity. On datasets with strict cross-feature constraints (Supermarket Sales, Online Purchases), Vine Copula achieves near-perfect KS fit ($> 96\%$) yet suffers 88%–100% dependency violation rates.

Dataset	Generator Architecture	Stat. Fidelity (KS ↑)	Sem. Integrity (HIF ↑)	Violation Rate (↓)
Supermarket Sales	Gaussian Copula	0.881	**0.972**	**2.4%**
	Vine Copula	**0.967**	0.020	100.0%
	CTGAN	0.799	0.045	98.8%
Online Purchases	Gaussian Copula	0.795	**0.692**	**29.4%**
	Vine Copula	**0.970**	0.250	88.0%
	CTGAN	0.659	0.291	86.0%
Credit Default	Gaussian Copula	0.744	**0.721**	**18.6%**
	Vine Copula	**0.963**	0.551	36.8%
	CTGAN	0.543	0.528	41.4%
Adult Income	Gaussian Copula	0.842	**0.762**	**19.6%**
	Vine Copula	**0.966**	0.754	19.8%
	CTGAN	0.747	0.698	25.4%

4.6 Qualitative Analysis: Relational Subtotal Violations

A qualitative audit of the dependency violations identified by HIF on transaction datasets (Online Purchases, Supermarket Sales) revealed systemic relational inconsistencies. In these records, generative models successfully match the marginal distribution of individual columns (`unit_price`, `quantity`, `subtotal`) in isolation, but fail to respect the arithmetic dependency subtotal = unit_price × quantity. Specifically, HIF identified synthetic records with positive subtotals where item quantity was zero, or where subtotals diverged significantly from the product of quantity and price. While standard KS tests and joint correlation metrics ignore these violations because each column value falls within its valid numeric range, HIF's LSE and NIC modules flagged them with maximum penalties, demonstrating the necessity of row-level conditional auditing.

4.7 Differential Diagnostic: Integrity vs. Loyalty

To determine whether HIF provides unique utility over joint statistical metrics, we evaluated its response at low corruption levels ($p \in [0.001, 0.1]$). We compared the behavior of the HIF score with the Joint Correlation Distance (JCD) to distinguish between inconsistency detection and drift detection.

Our results confirm that while the JCD acts as a high-sensitivity instrument for broad distributional drift, it lacks specificity to identify record-level conditional dependency violations. HIF decisively penalizes records with feature combinations that are individually plausible yet improbable under the learned joint distribution—violations that aggregate metrics dilute by averaging across all records. This provides a row-level dependency diagnostic that complements, rather than replaces, aggregate fidelity metrics: the two tools address fundamentally different failure modes.

5 Discussion

The results demonstrate that the Hybrid Integrity Framework provides a "strictly conservative" diagnosis. By employing a multiplicative aggregation model, HIF ensures that a failure in any single semantic dimension results in a low integrity score. This non-compensatory evaluation is critical for high-stakes domains, where a single inconsistent record can invalidate an analysis.

5.1 Reconciling Generative Paradigms: Statistical vs. Neural Trade-offs

A central finding of our cross-architecture audit (Table 5) is that the Dependency Gap is not restricted to deep neural black-boxes; rather, it is a universal pathology affecting both statistical and neural generative models in distinct ways.

Flexible statistical models, specifically Vine Copulas, achieve near-perfect per-column marginal fidelity (KS ≥ 0.963). However, because sequential bivariate copula decompositions smooth over higher-order joint boundaries, they suffer catastrophic dependency failure (HIF ≤ 0.250, violation rates 88.0%–100.0%) when auditing datasets with strict cross-feature relational constraints (Supermarket Sales, Online Purchases). Conversely, neural models such as CTGAN leverage adversarial representations that capture non-linear categorical dependencies better than rigid parametric copulas, yet still produce localized joint violations due to mode fragmentation.

This reconciles our core motivation: the Dependency Gap does not stem from neural opacity alone, nor from copula rigidity. It is an inherent artifact of standard generative loss functions, which optimize aggregate distributional fit (likelihood, Wasserstein distance, or marginal discrimination) rather than row-level joint conditional integrity. Consequently, post-hoc row-level auditing is necessary regardless of whether the underlying generator is statistical or neural.

5.2 Dependency Hub Selection via Feature Predictability

A current limitation of the Logical Sentinel Ensemble (LSE) is its reliance on single-variable classification accuracy ($S(h)$) for hub discovery, which primarily captures categorical dependencies. While computationally efficient, it may under-represent higher-order parity relationships. Future work will investigate Total Correlation and multi-way interaction kernels to broaden hub selection capability.

5.3 Heuristic Robustness and the Variance of Inclusion

We acknowledge that the use of a fixed 5th-percentile Confidence Floor (δ_h) is a heuristic calibrated for datasets with sufficient mass. As demonstrated in our hyper-parameter sensitivity sweep (Appendix E), the framework is robust to choices between the 1st and 5th percentiles. However, the empirical percentile may become unstable in extremely sparse or long-tailed distributions, such as those found in fraud detection and rare disease cohorts. For such regimes, we propose the adoption of Laplacian-smoothed percentiles or adaptive floors that scale with the local manifold density. This ensures that the “polygraph” remains robust, even when auditing the most imbalanced semantic domains.

5.4 Practical Interpretation and Computational Scalability

Finally, we address the interpretation of conservative integrity scores. Because HIF computes the multiplicative product across L auditing layers, the overall score $H(\mathbf{x}) = \prod_l \mathcal{C}_l$ requires simultaneous joint satisfaction of all conditional models. This conservative behavior is an intentional design choice for high-stakes auditing where severe failure on any single conditional model should penalize the composite score.

The LSE architecture is highly scalable, with a computational complexity of $O(N \cdot K)$ for auditing N records across K hubs. The parallelizable nature of this design allows it to be scaled effectively to tables with thousands of columns. Crucially, our empirical sweep (Appendix E) reveals that performance remains completely stable even at $K = 1$, meaning the hub count can be safely minimized to practically eliminate the K-multiplier overhead. In our benchmarks, auditing 10,000 synthetic rows required less than 15 s on standard 8-core commodity hardware. However, the memory overhead associated with training and storing an ensemble of K Random Forest sentinels for ultra-high-dimensional tables (e.g., $D > 10,000$) remains a potential hardware bottleneck. In such regimes, we propose replacing complex forest-based sentinels with sparse-linear or neuro-inspired models that offer a better trade-off between semantic depth and memory usage than existing models.

5.5 Specificity-Sensitivity Trade-off

A critical observation during our validation was the relative performance of HIF compared to global statistical metrics (e.g., JCD) under random corruption. While JCD exhibits higher sensitivity to broad marginal distributional drift, HIF prioritizes *dependency specificity*: it is calibrated to flag violations of learned conditional relationships, not marginal distributional jitter. This is a deliberate design choice. A generic anomaly detector sensitive to marginal outliers (such as LOF) would produce false positives on valid but rare records; HIF’s sensitivity is bounded by the confidence floors learned from the real data distribution. This specificity–sensitivity trade-off is essential for high-stakes applications, where a dataset’s utility depends on the preservation of inter-feature dependencies rather than the suppression of all marginal variation.

5.6 Broader Impact

By providing a verifiable “Integrity Certificate,” HIF strengthens the case for reproducible research in privacy-sensitive domains. We recommend using the HIF primarily as a diagnostic oracle to guide generative model refinement rather than as a purely exclusionary filter.

Quantitative Privacy Audit.

To address the privacy implications of integrity filtering, we conducted a Membership Inference Attack (MIA) audit using distance-to-closest-record (DCR) shadow modeling in the validation pipeline. Across the reported runs, the resulting AUC values remained near random-guessing behavior ($AUC = 0.491 \pm 0.01$), indicating no clear membership leakage signal introduced by integrity filtering in this setup, even when measured against the stringent standards of modern differential privacy [24].

6 Conclusion

The generation of synthetic tabular data is pivotal for democratizing research in highly regulated sectors. However, as generative models grow in complexity, the metrics used to evaluate them must evolve beyond aggregate marginal alignment to assess row-level joint distributional consistency. In this study, we introduced the Hybrid Integrity Framework (HIF), a probabilistic diagnostic for detecting inter-feature dependency violations in synthetic tabular data. By combining the Logical Sentinel Ensemble (LSE), which learns conditional feature dependencies via discriminative classifiers, with Neighbor-Invariant Continuity (NIC), which scores continuous feature plausibility within the learned categorical context, HIF provides a technically rigorous foundation for identifying the "Dependency Frontier" of a synthetic cohort.

Our empirical results on Census ACS and transaction benchmarks demonstrate that HIF achieves perfect Spearman monotonicity with respect to targeted dependency corruption—while aggregate metrics such as JCD and KS remain completely static under the same corruption, confirming that they evaluate different failure modes. Our cross-architecture audit reveals a persistent "Integrity-Fidelity Decoupling": sophisticated models such as Vine Copulas can achieve near-perfect distributional fidelity (KS > 0.96) while failing over 88% of dependency integrity tests on constrained datasets. By implementing non-compensatory probabilistic aggregation across independently learned dependency models, HIF provides researchers with a conservative, row-level certificate of joint distributional consistency—specifically, of conformity to the conditional feature dependencies of the real data—that aggregate marginal metrics cannot capture. We further show that HIF outperforms standard geometric outlier detectors (Isolation Forest, LOF) on conditional dependency violations while remaining appropriately insensitive to marginal noise types that do not constitute dependency failures. Future work will extend this framework to high-frequency financial time-series data [25] and explore kernel-based projections to capture higher-order dependencies [26–28]. Ultimately, HIF complements aggregate fidelity metrics by providing a specialized row-level dependency diagnostic, enabling more reliable synthetic datasets for high-stakes scientific research.

Code and Data Availability

The complete open-source implementation, experimental benchmarks, and dataset configurations are available anonymously at https://anonymous.4open.science/r/tabular-polygraph-21CE.

Acknowledgements

This research was supported by the Department of Data Science and Engineering, Eötvös Loránd University, Budapest, Hungary. The authors would like to thank the open-source community for the foundational tools used to develop the Hybrid Integrity Framework (HIF).

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A Formal Guarantees for the HIF Score

We provide a formal analysis of the Hybrid Integrity Framework (HIF) score properties under the **structured drift-proof (SDP)** framework introduced by Guo et al. [10].

A.1 Proof of Intersection Soundness

Theorem 1. The HIF score $H(\mathbf{x})$ defines a membership certificate for the intersection of semantic manifolds $\mathcal{M}_{total} = \bigcap_l \mathcal{M}_l$.

$$H(\mathbf{x}) \leq \prod_l \mathbb{1}[\mathbf{x} \in \mathcal{M}_l] \tag{10}$$

where L is the number of auditing layers in the model. By definition, $\mathcal{C}_l(\mathbf{x}) = 1$ if $\mathbf{x} \in \mathcal{M}_l$ and $\mathcal{C}_l(\mathbf{x}) \in [0, 1)$ otherwise.

1. **Completeness:** If $\mathbf{x} \in \mathcal{M}_{total}$, then $\mathbf{x} \in \mathcal{M}_l$ for all l. Thus, $\mathcal{C}_l(\mathbf{x}) = 1$ for all l, and $H(\mathbf{x}) = \prod_l 1 = 1$.
2. **Soundness (Non-Compensatory Property):** Suppose $\mathbf{x} \notin \mathcal{M}_{total}$. Then, there exists at least one l^* such that $\mathbf{x} \notin \mathcal{M}_{l^*}$, implying $\mathcal{C}_{l^*}(\mathbf{x}) < 1$. Because $0 \leq \mathcal{C}_l(\mathbf{x}) \leq 1$ for all l, the multiplicative product $H(\mathbf{x}) = \mathcal{C}_{l^*}(\mathbf{x}) \cdot \prod_{l \neq l^*} \mathcal{C}_l(\mathbf{x}) < 1$.
3. **Strict Convergence:** If $\mathbf{x}$ exhibits an "Absolute Violation" in any layer ($\mathcal{P}_{l^*} \to 1 \implies \mathcal{C}_{l^*} \to 0$), the product $H(\mathbf{x}) \to 0$ regardless of the performance in other layers.

This confirms that $H(\mathbf{x})$ provides a strictly conservative diagnosis for the joint satisfaction of all latent manifold law.

A.2 Proof of Asymptotic Statistical Convergence

Theorem 2 (Asymptotic Convergence of the HIF Penalty). Let $\hat{P}_N(x_h \mid \mathbf{x}_{\setminus h})$ be the empirical conditional probability estimator (the Sentinel f_h) trained on a ground-truth dataset $\mathcal{D}_{real}$ of size N. Assuming the base estimator is statistically consistent (e.g., consistent non-parametric trees), as $N \to \infty$, $\hat{P}_N$ converges in probability to the true conditional distribution $P^*(x_h \mid \mathbf{x}_{\setminus h})$. Consequently, the empirical penalty $\mathcal{P}_h^{(N)}(\mathbf{x})$ converges to the true underlying dependency violation penalty.

$$\lim_{N\to\infty} P\left(\sup_{\mathbf{x}} \left|\mathcal{P}_h^{(N)}(\mathbf{x}) - \mathcal{P}_h^*(\mathbf{x})\right| > \epsilon\right) = 0 \quad \forall \epsilon > 0 \tag{11}$$

1. **Consistent Density Estimation:** Because the LSE leverages consistent non-parametric estimators (Random Forests), the empirical conditional probabilities uniformly converge to the true Bayesian posterior P^*, provided the feature space is bounded and the leaf nodes grow sub-linearly with N.
2. **Bounded False Positive Rate:** By defining the confidence floor δ_h as the empirical 5th-percentile of $\hat{P}_N$ on the training data, we ensure that as $N \to \infty$, the asymptotic false positive rate (the probability of penalizing a true ground-truth record) is strictly bounded above by $\alpha = 0.05$.
3. **Asymptotic Soundness:** Therefore, any synthetic record $\mathbf{x}_{syn}$ that is penalized by the converged HIF ($\mathcal{P}^* > 0$) is guaranteed to possess a conditional probability strictly lower than 95% of the true data manifold, providing a statistically rigorous guarantee that the record is a structural hallucination.

This provides the necessary statistical convergence guarantee, ensuring that HIF's empirical penalties are not arbitrary heuristics but asymptotically converge to true joint-distribution outlier bounds.

B Logical Sentinel Ensemble (LSE) Implementation Details

The LSE architecture prioritizes categorical dependency modeling by identifying and enforcing high-dependency hubs across categorical features.

B.1 Feature Dependency Hub Discovery

We ranked features based on their **Feature Dependency Score** $S(h)$, which measures the classification accuracy of predicting a feature within the full dataset context (including binned numeric features). In our experiments on the U.S. Census ACS dataset, the LSE identified the following top-5 hubs: `household_income`, `housing_cost`, `poverty_status`, `education`, and `age_group`.

B.2 Confidence Floor Calibration

The confidence floor δ_h is defined as the 5th-percentile of the probability mass that a Sentinel f_h assigns to the correct category in the ground-truth data $\mathcal{D}_{real}$, with a safety floor of 0.01:

$$\delta_h = \max(\text{Percentile}_5\left(\{f_h(\mathbf{x}_h \mid \mathbf{x}_{\setminus h}) \mid \mathbf{x} \in \mathcal{D}_{real}\}\right), 0.01) \tag{12}$$

This robust approach avoids the instability of 1st-percentile thresholds in sparse data distributions and ensures sentinels remain sensitive to conditional dependency violations [6, 17].

C Extended Results and Sensitivity Analysis

C.1 Hyperparameter Sensitivity

We evaluated HIF robustness across three key hyperparameters on the Census ACS dataset (2,000 records with 5% injected violations). All experiments used $N = 5$ random seeds; we report mean ± SEM. The continuous HIF score is the primary metric; the violation rate is derived by applying the violation threshold to row-level penalties.

C.1.1 Hub Count K

Table 6 shows that the HIF score is stable across all hub counts, confirming that the framework's integrity signal is not dependent on a specific number of dependency hubs. The geometric mean aggregation ensures that adding more sentinels does not dilute the score.

Table 6 Sensitivity to hub count K (percentile=5, threshold=0.5).

K	HIF Score (±SEM)	Violation Rate (±SEM)
1	0.9284 ± 0.0004	0.036 ± 0.001
3	0.9284 ± 0.0004	0.036 ± 0.001
5	0.9284 ± 0.0004	0.036 ± 0.001
10	0.9284 ± 0.0004	0.036 ± 0.001
15	0.9284 ± 0.0004	0.036 ± 0.001

C.1.2 Confidence Floor Percentile

Table 7 shows that the HIF score is stable for percentiles 1–5, with a minor 0.2% drop at the 10th percentile. The 5th-percentile default provides the optimal balance between sensitivity to genuine violations and robustness against borderline records.

C.1.3 Violation Threshold

Table 8 demonstrates that the continuous HIF score is invariant to the binary classification threshold, as expected. The violation rate scales monotonically with the threshold, allowing practitioners to tune the operating point for their specific precision–recall requirements without retraining.

Table 7 Sensitivity to confidence floor percentile ($K = 5$, threshold=0.5).

Percentile	HIF Score (±SEM)	Violation Rate (±SEM)
1	0.9284 ± 0.0004	0.036 ± 0.001
3	0.9284 ± 0.0004	0.036 ± 0.001
5	0.9284 ± 0.0004	0.036 ± 0.001
10	0.9260 ± 0.0003	0.038 ± 0.000

Table 8 Sensitivity to violation threshold ($K = 5$, percentile=5).

Threshold	HIF Score (±SEM)	Violation Rate (±SEM)
0.3	0.9284 ± 0.0004	0.076 ± 0.000
0.5	0.9284 ± 0.0004	0.036 ± 0.001
0.7	0.9284 ± 0.0004	0.023 ± 0.000

C.2 Scalability and Baseline Comparisons

We benchmarked HIF against the **SHAP Distance** semantic fidelity metric [9], which serves as a state-of-the-art baseline for explainability-aware evaluation. Although the SHAP Distance effectively identifies semantic drift, it suffers from severe scalability bottlenecks. Auditing N records with HIF scales linearly, $O(N \cdot K)$, where K is the number of hubs. In our empirical trials, auditing 100,000 synthetic records with LSE (5 hubs) took ≈ 1.2 min on a single CPU core. In contrast, SHAP-based methods require $O(N \cdot D \cdot 2^D)$ operations or expensive approximations, taking several hours for the same volume of data. Thus, HIF provides comparable detection accuracy while maintaining the necessary computational efficiency to scale to high-cardinality datasets with millions of rows of data.

C.3 Privacy-Fidelity Trade-off (TAMIS)

Using the TAMIS Privacy Oracle, we confirmed that the HIF-filtered records did not increase membership inference vulnerability. In fact, by removing outlier records that often serve as unique linkage fingerprints, HIF filtering marginally improves the privacy profile of synthetic cohorts.

C.4 Cross-Architecture Violation Frontiers

Our Cross-Architecture Maturity Audit revealed distinct "Violation Frontiers" for different generative families. While statistical models (**Gaussian and Vine Copulas**) tend to violate constraints through excessive smoothing (averaging out manifold sharp edges), neural models like **CTGAN** exhibit *Combinatorial Violations*.

Specifically, CTGAN often fragments the latent manifold during adversarial training, leading to synthetic records that satisfy the marginal distributions for each feature but inhabit “Impossible Junctions”—categorical combinations that have zero probability mass in the ground-truth manifold. The Logical Sentinel Ensemble (LSE) is particularly effective at identifying these junctions, as the sentinels assign near-zero probability to the violated categories conditioned on their manifold hubs. This architectural discrepancy explains the higher base integrity score of CTGAN ($H = 0.382$) compared to that of Gaussian Copulas ($H = 0.203$), despite the latter's simpler statistical structure.

C.5 Macro-Economic Violation Detection

The application of HIF to the **World Bank Development Indicators** ($D = 12$) highlights the framework's effectiveness in macro-level auditing. Macroeconomic data are governed by strict structural relationships (e.g., the relationship between GDP growth, inflation, and the Gini index).

As shown in our audit, all evaluated generative models suffered from a significant “Integrity Deficit” in the World Bank domain, with integrity scores dropping near $H = 0.00$ in several trials. This suggests that for macroeconomic indicators governed by global state-level laws, standard generative architectures fail to recover sparse semantic manifolds. By leveraging the **Neighbor-Invariant Continuity (NIC)** module, the HIF correctly identifies these records as continuous manifold violations, even when the distributional fidelity appears to be high.

D Limitations and Open Challenges

While the Hybrid Integrity Framework (HIF) provides a technically rigorous diagnostic for synthetic data, our current formulation faces three key challenges identified through empirical auditing.

D.1 Dependency Hub Selection via Feature Predictability

The Logical Sentinel Ensemble (LSE) relies on single-variable classification accuracy ($S(h)$) to discover dependency hubs. Although computationally efficient, it primarily captures categorical dependencies. Future work will investigate **Total Correlation** and multi-way interaction kernels to generalize hub selection to complex parity regimes.

D.2 Sensitivity on Sparse or Imbalanced Manifolds

The use of the **5th-percentile Confidence Floor** (δ_h) is a data-driven heuristic designed for stability. However, on extremely sparse or highly imbalanced datasets (e.g., rare fraud events or sparse healthcare codes), the calibration of δ_h may be susceptible to outliers in ground-truth data. In such cases, the framework may require domain-specific tuning of the percentile parameter or the adoption of Laplacian smoothing to stabilize the boundary conditions.

The parallelizable nature of the LSE architecture allows it to scale effectively to tables with thousands of columns. However, the memory overhead associated with training and storing an ensemble of K Random Forest sentinels for ultra-high-dimensional tables (e.g., $D > 10,000$) remains a potential hardware bottleneck. In such regimes, we propose replacing complex forest-based sentinels with sparse-linear or neuro-inspired models that offer a better trade-off between semantic depth and memory usage.

E Hyperparameter Sensitivity

To validate the robustness of HIF against manual configuration, we performed a sweep across three core hyperparameters on the Census ACS dataset ($N = 10$ seeds): the number of categorical hubs (K), the confidence percentile floor (δ_h), and the final violation threshold. As shown in Table 9, the framework is remarkably insensitive to these parameters.

Specifically, the integrity score and violation rate remain virtually identical across all $K \in [1, 15]$, demonstrating that a small number of sentinel hubs is sufficient to capture the structural manifold. This empirical stability justifies setting K to a small integer, drastically reducing the $O(N \cdot K)$ computational overhead discussed in Section 5.4. Similarly, the 5th-percentile confidence floor provides stable rejection rates, confirming the heuristic robustness discussed in Section 5.3.

Table 9 Hyperparameter Sensitivity on Census ACS ($N = 10$ seeds). The framework exhibits extreme stability across hub counts (K) and percentiles. Changing the violation threshold predictably alters the rejection rate without destabilizing the mean integrity score.

Parameter	**Value**	**HIF Score (mean)**	**Violation Rate**
Hub Count (K)	1	0.928	3.6%
	3	0.928	3.6%
	5	0.928	3.6%
	10	0.928	3.6%
	15	0.928	3.6%
Confidence Percentile (δ_h)	1st	0.928	3.6%
	3rd	0.928	3.6%
	5th	0.928	3.6%
	10th	0.926	3.8%
Violation Threshold	0.3	0.928	7.6%
	0.5	0.928	3.6%
	0.7	0.928	2.3%